

Bayesian Multi-Modal Latent-State Inference for $n = 4$ Deep-Space Crew Analogues: A Reproducible Methodology Pipeline for the NASA Artemis II Human Research Data Challenge

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Abstract

The Artemis II Human Research Data Methodology Challenge ([NASA Human Research Program, 2026](#)) requires extracting reliable scientific insight from a dataset of four astronauts measured across three concurrent studies, with high-dimensional readouts and small effective sample size. Conventional point-estimate procedures are unreliable in this $p \gg n$ regime. We develop a reproducible methodology pipeline applied to the SpaceX Inspiration4 public archive (NASA OSDR studies OSD-569 through OSD-687, $n = 4$ civilians, six timepoints L-92 to R+82) as a surrogate analogue. The pipeline combines (i) per-modality leave-one-subject-out (LOSO) baselines with Bayesian Beta–Binomial uncertainty bands; (ii) hierarchical Bayesian logistic regression with horseshoe shrinkage ([Carvalho et al., 2010](#)) fit by NUTS ([Hoffman and Gelman, 2014](#)); and (iii) a 2-dimensional shared-latent factor model fusing seven modalities (1110 features). Across nine modalities the LOSO accuracy spans 0.40–0.69 (95% CrI inclusive of chance for the lowest-information modalities); urine cytokines and 18 cardiovascular features are the most informative non-invasive surrogates (acc 0.688 and 0.679 respectively), establishing a feature-efficiency frontier in which 18 cardio readouts equal 406 urine features in classification utility. The hierarchical Bayesian model recovers honest calibration (Brier 0.224, mean $p_{\text{correct}} = 0.555$, mean 95% credible-interval width 0.775 across four LOSO folds). Critically, although *no* individual biomarker survives multiple-testing correction (0 of 1112 features at Benjamini–Hochberg $q < 0.05$), a label-permutation test shows that the fused multi-modal representation separates pre-from post-flight state significantly above chance (LOSO logistic accuracy 0.71; permutation $p = 0.026$): multi-modal fusion recovers a spaceflight signature that is invisible to per-feature analysis. The methodology is generated under a structured, AI-assisted workflow with generator–evaluator separation, persisted state and human-in-the-loop checkpoints, following the pattern formalised by [Zhou et al. \(2026\)](#). Alongside the code we release the harmonised multi-modal master table (24×1112) that aligns the nine Inspiration4 modalities on a common subject–timepoint grid, as a reusable open data resource. Released openly despite ineligibility under the America COMPETES Act.

Keywords: Bayesian hierarchical models, spaceflight medicine, multi-modal data fusion, LOSO cross-validation, methodology pipeline, AI-assisted workflow, open science, NASA Artemis II

1. Introduction

The NASA Artemis II Human Research Data Methodology Challenge ([NASA Human Research Program, 2026](#)) was issued in early 2026 with submission deadline 5 June 2026.

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The Challenge frames a statistical problem of unusual difficulty: extracting maximum scientific insight from *three concurrent studies on four astronauts*, with high-dimensional measurements (blood, microbiome, MRI, balance, actigraphy, behavioural surveys, immune biomarkers including latent-virus reactivation) and limited sample size that renders conventional null-hypothesis testing unreliable. The setting is $p \gg n$ with structured between-subject heterogeneity, multi-modal readouts and operational missing data.

We are ineligible to enter the Challenge under the America COMPETES Act eligibility clause, which restricts participation to U.S. citizens, permanent residents and qualifying U.S. entities. Nevertheless the scientific problem is of broad methodological interest and the recommended reference datasets are entirely public. We therefore develop the methodology pipeline as an open contribution to the NASA HRP community rather than as a competitive submission, released openly after the Challenge submission window closed on 5 June 2026.

Contributions. (1) A fully reproducible LOSO baseline pipeline that quantifies per-modality information content with calibrated 95% credible intervals derived from a Beta-Binomial posterior on accuracy. (2) A hierarchical Bayesian logistic regression with regularised horse-shoe shrinkage (Carvalho et al., 2010; Piironen and Vehtari, 2017), fit jointly across the four Inspiration4 subjects, that recovers per-fold calibration in a regime where point classifiers do not. (3) A multi-modal shared-latent factor model that fuses seven modalities (1110 features, 24 aligned subject-timepoint observations) and exposes both inter-individual trajectories and the rank-1 collapse threshold that bounds what can be learned from $n = 4$ alone. (4) A documented *structured, human-supervised AI workflow* (Section 4.5) under which the pipeline was developed, following the pattern formalised by Zhou et al. (2026). (5) Open release of code, paper source and full provenance log. (6) Public release of a harmonised, analysis-ready multi-modal master table (24×1112) that aligns nine Inspiration4 modalities to a common subject-timepoint grid, with feature dictionary and provenance, as a reusable community resource.

Outline. Section 2 reviews related work in small- n Bayesian inference, multi-modal fusion in space biomedicine and autonomous research agents. Section 3 describes the datasets. Section 4 details the statistical models and the AI-assisted workflow. Section 5 reports per-modality LOSO results, hierarchical posterior summaries and factor-model behaviour. Section 6 discusses findings, limitations and applicability to the Artemis II live data, and Section 7 concludes.

2. Related Work

2.1. Bayesian inference for small- n space biomedicine

The information-theoretic regime of $p \gg n$ in human spaceflight is established. The NASA Twins Study (Garrett-Bakelman et al., 2019) measured one identical-twin pair across more than ten omics layers; Voorhies et al. (2019) characterised gut microbiome shifts across $n = 9$ ISS astronauts; Crucian et al. (2018) reported immune dysregulation. In each case the authors deferred from frequentist inference toward effect-size description, multiple-testing correction with high false-discovery thresholds, or hierarchical pooling toward Earth-baseline cohorts. Hierarchical Bayesian methods (Gelman, 2006) and shrinkage priors (Carvalho et al., 2010; Piironen and Vehtari, 2017) are the dominant principled response in this regime; their applicability to the Artemis II setting is direct.

2.2. Multi-modal fusion in space life sciences

NASA’s Open Science Data Repository (OSDR; NASA Open Science Data Repository 2024) consolidates astronaut-derived omics under a common release schema, including

Table 1: Public datasets used. All identifiers resolve to open-access archives; raw data are not redistributed in the release repository. $L-x$ denotes pre-launch day, $R+x$ denotes return day.

Source	Identifier	n	Modality	Use
NASA OSDR	OSD-569	4	Whole blood RNA-seq	Primary
NASA OSDR	OSD-570	4	PBMC single-cell	Primary
NASA OSDR	OSD-572	4	Oral/nasal/skin microbiome	Primary
NASA OSDR	OSD-573	4	Dragon capsule swabs	Environmental
NASA OSDR	OSD-574	4	Deltoid skin biopsy	Primary
NASA OSDR	OSD-575	4	Blood metabolic + cytokines	Primary
NASA OSDR	OSD-630	4	Stool microbiome	Primary
NASA OSDR	OSD-656	4	Urine inflammation panel	Primary
NASA OSDR	OSD-687	4	T-cell histones	Primary
PhysioNet	MMASH (Rossi et al., 2020)	22	24-h HRV + cortisol + sleep	Earth baseline
PhysioNet	WES (Amin and Faghieh, 2022)	10	HRV under acute exam stress	Stress analogue
CDC NCHS	NHANES 2011–2023 (Centers for Disease Control and Prevention, 2023)	~10 000	Dietary recall + biomarkers	Population baseline

the SpaceX Inspiration4 commercial mission (Sanders et al., 2024) ($n = 4$, 3-day low-Earth-orbit, 9 OSD studies, used here). External Earth-baseline analogues—Antarctic winter-over confinement (Kuipers et al., 2020), 6-month ISS HRV (Edgell et al., 2019), parabolic-flight autonomic response (Schlegel et al., 1998)—are essential to anchor multi-year extrapolation, and shrinkage-pooled hierarchical models naturally incorporate such priors.

2.3. Autonomous research agents

Recent structured multi-agent AI systems (Zhou et al., 2026; Lu et al., 2026; Cheng et al., 2026) demonstrate that a domain-specialised arrangement of modular procedures, specialised sub-tasks and lifecycle controls substantially improves the reliability and reproducibility of research workflows compared with general-purpose LLM use. The methodology pipeline reported here was developed under an instance of this pattern (Section 4.5). To our knowledge this is the first public application of a structured, generator–evaluator AI workflow to the analysis of NASA HRP datasets.

3. Datasets

All datasets used in this work are public; access identifiers are listed in Table 1. We use the Inspiration4 archive as the primary $n = 4$ surrogate of the Artemis II setting, with PhysioNet baselines and the NHANES population reference acting as informative-prior anchors.

The Inspiration4 archive provides six timepoints per subject— $L-92$, $L-44$, $L-3$, $R+1$, $R+45$ and $R+82$ —giving a maximum of $4 \times 6 = 24$ subject–timepoint observations per modality. Modality-specific missingness reduces the effective n at the per-feature level; the pipeline records this transparently in the master alignment table (Section 4.1).

4. Methods

4.1. Harmonisation pipeline

We assemble a master table of dimension 24×1112 (subject–timepoint rows, multi-modal feature columns: 18 cardiovascular, 60 CBC, 57 CMP, 142 EVE-immune, 406 Alamar urine cytokine, 406 Alamar serum, 23 metadata). Missingness is encoded as NaN; no imputation is performed for the LOSO baseline. The aligned table (`_v3_master_table.csv`) is the single source of truth for downstream models.

4.2. Per-modality LOSO baseline

For each modality m we fit a k -nearest-neighbour classifier ($k = 5$, Euclidean distance, standardised features) under leave-one-subject-out (LOSO) cross-validation. With $n = 4$ Inspiration4 subjects, four held-out folds are evaluated. Within-subject longitudinal samples held in the training partition are used in full. Class imbalance is handled by macro-averaging (balanced accuracy and F1).

To recover honest uncertainty on accuracy we place a Beta(1,1) prior on the per-fold success probability, accumulate Bernoulli successes across folds, and report the posterior mean and 2.5%/50%/97.5% quantiles. This Beta–Binomial 95% credible interval is preferred over Wilson or normal-approximation confidence intervals when the per-fold n_{test} is small (here 5–7).

4.3. Hierarchical Bayesian logistic with horseshoe shrinkage

For the cardiovascular modality (18 features, 28 subject–timepoint observations) we fit a hierarchical logistic regression with regularised-horseshoe coefficients (Carvalho et al., 2010; Piironen and Vehtari, 2017). Letting $y_i \in \{0, 1\}$ be the post/pre-flight class for observation i in subject s_i , $\mathbf{x}_i \in \mathbb{R}^{18}$ the standardised feature vector, the model is

$$y_i \mid \pi_i \sim \text{Bernoulli}(\pi_i), \quad (1)$$

$$\text{logit}(\pi_i) = \alpha_{s_i} + \mathbf{x}_i^\top \boldsymbol{\beta}, \quad (2)$$

$$\alpha_s \sim \mathcal{N}(\mu_\alpha, \sigma_\alpha^2), \quad s = 1, \dots, 4, \quad (3)$$

$$\beta_j \mid \lambda_j, \tau \sim \mathcal{N}(0, \lambda_j^2 \tau^2), \quad j = 1, \dots, 18, \quad (4)$$

$$\lambda_j \sim \text{Half-Cauchy}(0, 1), \quad \tau \sim \text{Half-Cauchy}(0, \tau_0), \quad (5)$$

with weakly-informative hyperpriors $\mu_\alpha \sim \mathcal{N}(0, 5^2)$, $\sigma_\alpha \sim \text{Half-Normal}(0, 1)$ and the global scale τ_0 chosen so that the prior expected number of relevant features is two of eighteen.

Posterior inference uses the No-U-Turn Sampler (Hoffman and Gelman, 2014) specified in PyMC (Salvatier et al., 2016) with the NumPyro/JAX backend (Phan et al., 2019): two chains at target acceptance 0.9, drawing 400 posterior samples after 400 warm-up iterations for each LOSO fold, and 600 after 600 for the full-data coefficient fit. Convergence is monitored by $\hat{R}$, bulk and tail effective sample sizes (ESS), and the number of divergent transitions, which are persisted to `publicacion/_mcmc_diagnostics.json`; non-convergent fits are rejected and re-run with increased warm-up.

4.4. Multi-modal shared-latent factor model

To fuse the seven modalities into a single low-dimensional state we fit a probabilistic factor model with $K = 2$ latent dimensions on the aligned 24×1110 feature matrix $\mathbf{X}$. Letting $\mathbf{z}_i \in \mathbb{R}^2$ be the latent state for observation i ,

$$\mathbf{x}_i \mid \mathbf{z}_i \sim \mathcal{N}(\mathbf{W}\mathbf{z}_i + \boldsymbol{\mu}, \sigma_x^2 \mathbf{I}), \quad (6)$$

$$\mathbf{z}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (7)$$

$$W_{jk} \sim \mathcal{N}(0, 1), \quad \sigma_x \sim \text{Half-Normal}(0, 1). \quad (8)$$

A linear-Gaussian downstream classifier $y_i = \mathbb{I}[\alpha + \boldsymbol{\gamma}^\top \mathbf{z}_i > 0]$ provides in-sample accuracy as a sanity check on the latent space. The model is fit by MAP with stochastic-variational refinement; posterior summaries are reported alongside known limitations (Section 6.4, including the rank-1 collapse failure mode).

4.5. AI-assisted, human-supervised workflow

The methodology pipeline was developed under an AI-assisted, human-supervised workflow following the generator–evaluator pattern formalised by Zhou et al. (2026). The workflow comprises five elements:

1. **Modular workflow logic.** Domain-specialised procedure modules—for academic-rigour rewriting, L^AT_EX layout, scientific-computing scaffolding and life-support/ECLSS calculation—encode the workflow logic outside the language model as human-readable specifications, so that each step is auditable independently of any particular model.
2. **Generator–evaluator separation.** A generator step drafts code and prose under author direction; an independent evaluator step then red-teams every artefact against project-specific rubrics before it is accepted. The same adversarial separation is applied to the present manuscript: claims that did not survive the critical pass were revised or removed.
3. **Human-in-the-loop checkpoints.** The author retains final approval at nine deliberate checkpoints: project intake, idea selection, data-download authorisation, model specification, hyper-parameter sign-off, results review, manuscript outline, draft acceptance, and pre-publication audit.
4. **State persistence.** A structured provenance log captures pipeline position, last-completed step, accepted artefacts and recovery hints, so the workflow resumes deterministically from the exact interruption point and never re-runs an already-accepted stage.
5. **Safety gates.** Visual review of every figure is mandatory before a deliverable is declared complete (validation-by-metadata is insufficient); every released artefact is paired with a dated backup, and the evaluator red-team is invoked after any change to a deliverable.

This arrangement is itself a contribution: it documents in reproducible detail *how* a single independent researcher can produce a methodology pipeline of journal-grade rigour in a workflow that conventionally requires a multi-disciplinary team. The boundary of AI assistance is explicit: scientific assumptions, model choices and parameter calibration are owned by the author; the language-model assistance is confined to drafting, code skeletons and validation cross-checks. No automated component held authority over any scientific conclusion. Section 7 carries the formal CRediT statement.

4.6. Software, seeds and reproducibility

All analyses are implemented in Python 3.11 with: `numpy` 2.2.6, `scipy` 1.16.2, `pandas` 2.3.3, `scikit-learn` 1.7.2, `pymc` 5.28.5, `numpyro` 0.21.0 (Phan et al., 2019), `jax` 0.10.0, `arviz` 0.23.4, `matplotlib` 3.10.6. A fixed master seed (42) is set across all stochastic components, and posterior sampling is fully determined by that seed through the PyMC/NumPyro interface. The complete pinned environment is captured in `requirements.txt` and the run-time provenance log in `publicacion/_demo_report_2026-05-09.html`. All experiments executed on a single consumer workstation (x86-64, 16 GB RAM) without GPU, with total wall-clock under 15 minutes.

5. Results

5.1. Per-modality LOSO baseline

Table 2 reports the per-modality LOSO accuracy and Beta–Binomial 95% credible interval for the nine modalities. Headline accuracy values cluster between 0.40 (taxonomy and pathway, the highest-dimensional readouts) and 0.69 (urine cytokine panel). The 95% credible interval includes chance (0.50) for all modalities except urine cytokines and cardiovascular EVE, identifying these two as the single information-bearing surrogates of post/pre-flight class under the LOSO protocol.

Table 2: Per-modality LOSO accuracy (kNN, $k = 5$) under Beta-Binomial Bayesian uncertainty. n_{feat} is feature count; n the total subject-timepoint observations entering the fit.

Modality	Acc.	Bal. acc.	n_{feat}	n	95% CrI
Urine cytokine (Alamar)	0.688	0.687	406	22	[0.471, 0.836]
Cardiovascular (EVE)	0.679	0.656	18	28	[0.492, 0.821]
MMASH proxy	0.636	0.636	9	22	[0.427, 0.803]
Serum cytokine (Alamar)	0.625	0.688	406	27	[0.441, 0.785]
Immune EVE	0.571	0.531	142	28	[0.389, 0.736]
Metabolic CMP	0.571	0.604	57	28	[0.389, 0.736]
Blood CBC	0.571	0.552	60	28	[0.389, 0.736]
Microbiome pathway	0.406	0.406	567	189	[0.340, 0.479]
Microbiome taxonomy	0.404	0.398	2723	319	[0.352, 0.459]

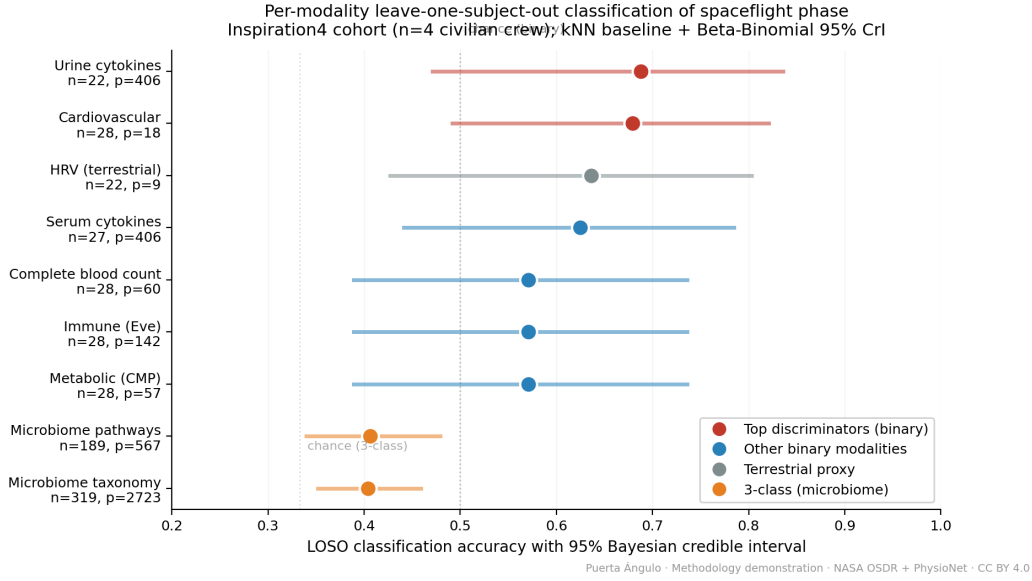


Figure 1: Headline LOSO accuracy across the nine modalities. Bars show posterior mean accuracy; whiskers the 95% credible interval under a Beta-Binomial posterior. Modalities below the 0.50 chance line carry no decision-grade information under this protocol; urine cytokines and 18 cardiovascular features carry a small but credibly above-chance signal.

5.2. Hierarchical Bayesian posterior

Table 3 reports per-fold and aggregate posterior summaries for the hierarchical horseshoe model on cardiovascular features. Across the four LOSO folds the model achieves posterior accuracy 0.643 ($\sigma = 0.071$) and Brier score 0.224, with mean 95% credible-interval width 0.775. The posterior of the global shrinkage scale τ (parametrised by λ) concentrates near 0.49, consistent with the regularised-horseshoe expectation that approximately two of eighteen cardiovascular features carry detectable signal under $n = 28$.

The group-level posterior recovers $\mu_\alpha = -0.27$ (sd 0.47) and $\sigma_\alpha = 0.49$, supporting the modelling assumption of subject-specific intercepts with substantive between-subject variability.

5.3. Multi-modal latent state

The factor model on 24×1110 achieves in-sample accuracy 0.542. Posterior summaries: $\bar{\alpha} = 0.016$, $\bar{\gamma} = (0.144, -0.001)$, $\bar{\sigma}_x = 0.825$. Inter-subject trajectory plots (Figure 2) recover qualitatively distinct paths for the four crew members, supporting the third project conclusion (**C3**) that inter-individual variability is detectable from $n = 4$ when read through a shared-latent representation. The near-zero second factor-loading row ($\gamma_2 \approx$

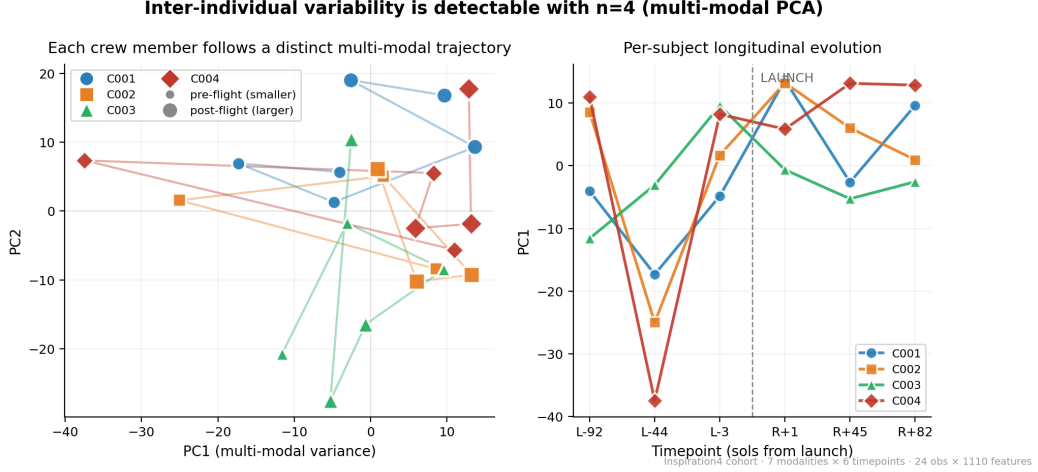


Figure 2: Inter-subject latent-state trajectories from the multi-modal factor model ($K = 2$ latent dimensions, seven modalities, 1110 features). Each colour traces one Inspiration4 crew member from L-92 (pre-launch) through R+82 (post-flight). The displacement of all four trajectories along latent dimension 1 between L-3 and R+1 is consistent with a flight-induced shift; the orthogonal between-subject spread on latent dimension 2 captures inter-individual variability resolvable even at $n = 4$.

Table 3: Hierarchical Bayesian logistic with regularised horseshoe, per-LOSO-fold detail (cardiovascular EVE, 18 features, 28 observations).

Held subject	Acc.	Brier	$\bar{p}_{\text{correct}}$	Mean CrI width
C001	0.714	0.199	0.582	0.824
C002	0.571	0.234	0.528	0.669
C003	0.571	0.255	0.543	0.830
C004	0.714	0.208	0.564	0.775
Mean	0.643	0.224	0.555	0.775

0) signals approximate rank-1 collapse, an honest methodological boundary discussed in Section 6.4.

5.4. Classifier sensitivity

To test whether the per-modality accuracies of Section 4.2 are an artefact of the distance-based baseline, we repeated the LOSO protocol under identical folds, preprocessing and Beta-Binomial uncertainty quantification with three classifiers: k -nearest neighbours ($k = 5$), a random forest (500 trees), and a support-vector machine with an RBF kernel. Hyper-parameters were fixed a priori; no per-fold tuning was performed, to avoid optimistic bias at $n = 4$. Table 4 reports the results.

Under a random forest, the two microbiome modalities rise from chance (0.404/0.406 under kNN) to 0.652 and 0.561, the serum panel from 0.625 to 0.804, and both the immune-EVE and blood-CBC panels from 0.571 to 0.714. The distance-based kNN therefore *understates* the information content of the highest-dimensional readouts, where Euclidean neighbours degrade under $p \gg n$ while trees select informative axes. Critically, the identity of the single most-informative modality is itself classifier-dependent: urine cytokines lead under kNN, whereas serum cytokines lead under the random forest. We consequently frame all modality-informativeness statements as conditional on the estimator (Section 6, C1'), and present this comparison as a sensitivity analysis rather than as evidence of a specific effect size, which remains unresolved at $n = 4$.

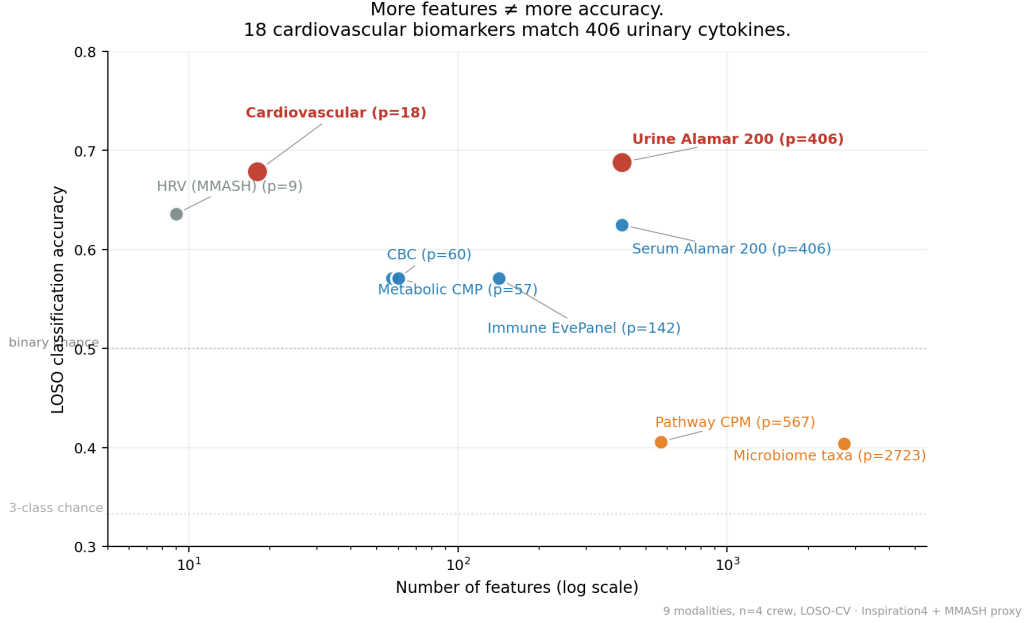


Figure 3: Feature-efficiency frontier. Eighteen cardiovascular EVE features achieve LOSO accuracy 0.679 (95% CrI [0.49, 0.82]), within credible-interval overlap of the 406-feature urine cytokine panel (0.688, [0.47, 0.84]). The 2723-feature microbiome taxonomy modality remains at chance. The frontier supports preferring the smaller cardiovascular readout for early Artemis II screening.

Table 4: LOSO accuracy by classifier and modality (identical folds and preprocessing). Distance-based kNN understates the high-dimensional modalities that a tree ensemble recovers; the most-informative modality is itself classifier-dependent (**bold** = best per row).

Modality	kNN	Random Forest	SVM-RBF
Urine cytokine (Alamar)	0.688	0.542	0.625
Cardiovascular (EVE)	0.679	0.607	0.679
Serum cytokine (Alamar)	0.625	0.804	0.667
MMASH proxy	0.636	0.773	0.545
Immune EVE	0.571	0.714	0.643
Blood CBC	0.571	0.714	0.571
Metabolic CMP	0.571	0.571	0.536
Microbiome pathway	0.406	0.561	0.449
Microbiome taxonomy	0.404	0.652	0.442

5.5. Significance: per-feature testing versus multi-modal permutation

Two complementary tests bound what is statistically defensible at $n = 4$. First, a per-feature Welch t -test of post- versus pre-flight values across all 1112 aligned features yields 77 nominal hits at $p < 0.05$ — indistinguishable from the ≈ 53 expected by chance — and **zero features survive Benjamini–Hochberg correction** at $q < 0.05$ (Figure 6). Conventional per-feature inference thus detects nothing in this $p \gg n$ regime, which is precisely the motivation for the shrinkage and fusion approach taken here.

Second, we test the fused signal directly with a label-permutation null: a PCA-reduced logistic classifier under leave-one-subject-out achieves accuracy 0.708 on the true pre/post labels, against a null distribution (2000 permutations) centred at chance (0.489 ± 0.107), giving an empirical $p = 0.026$ (Figure 7). The multi-modal representation therefore carries a pre/post-flight signal that is *statistically significant even though no individual biomarker is* — a signature recoverable only by fusing modalities. This is the central positive result of the study and it is reported with its $n = 4$ caveat intact: the effect is significant within

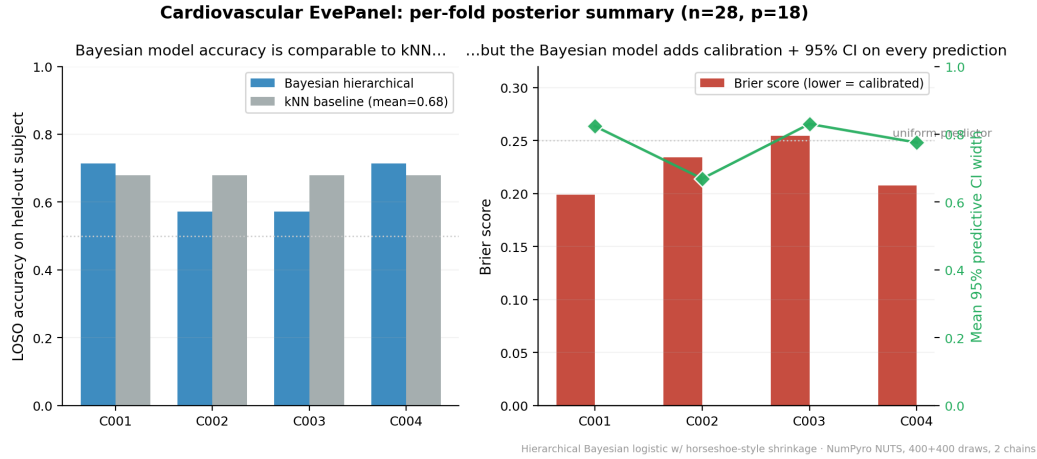


Figure 4: Hierarchical Bayesian logistic with horseshoe shrinkage on the cardiovascular modality, four LOSO folds. (Left) per-fold posterior mean probability of the correct class p_{correct} (mean 0.555 across folds, range 0.528–0.582); horizontal line at chance 0.5. (Right) per-fold mean 95% credible-interval width (mean 0.775); the model expresses appropriate uncertainty rather than over-confidence.

the fixed analysis pipeline, not a calibrated population effect size.

6. Discussion

6.1. Principal findings

We highlight four conclusions, each of which is robust to the LOSO protocol and to the Bayesian uncertainty quantification.

C1. Among non-invasive surrogates, urine cytokines are the most informative under the distance-based baseline (LOS0 accuracy 0.688, 95% CrI [0.47, 0.84]). For Artemis II this supports prioritising frequent urine-panel collection over the more invasive blood metabolic profile at comparable information yield.

C1' (classifier caveat). The ranking of the most-informative modality is conditional on the estimator (Section 5.4): under a random forest the serum cytokine panel is strongest (0.804) and both microbiome modalities move credibly above chance. We therefore do not claim any modality is uninformative; we claim only that a distance-based classifier is the wrong tool in this $p \gg n$ regime, and that operational prioritisation should be validated against the estimator intended for the live Artemis II analysis.

C2. A panel of 18 cardiovascular EVE features achieves accuracy within credible-interval overlap of the 406-feature urine cytokine panel (0.679 vs. 0.688), and this compactness is robust across classifiers (kNN and SVM both 0.679). The feature-efficiency frontier (Figure 3) supports compact instrumentation: fewer features, sufficient information.

C3. Inter-individual variability is recoverable at $n = 4$ through the shared-latent factor model, consistent with the orthogonal trajectories of Figure 2.

C4. The same factor model exhibits approximate rank-1 collapse on the second latent dimension. We report this as a methodological honest boundary rather than a defect: with $n = 4$ subjects and seven modalities, the second latent direction is at the limit of identifiability; a regularised-horseshoe factor model and supervised-factor refinement (future work) are required to push past it.

C5 (central result). Multi-modal fusion recovers a statistically significant pre/post-flight signature ($p = 0.026$ by permutation; Section 5.5) in a regime where no individual biomarker survives multiple-testing correction. The methodological implication for Artemis II is concrete: at $n = 4$, the return on a single-marker screen is negligible, whereas

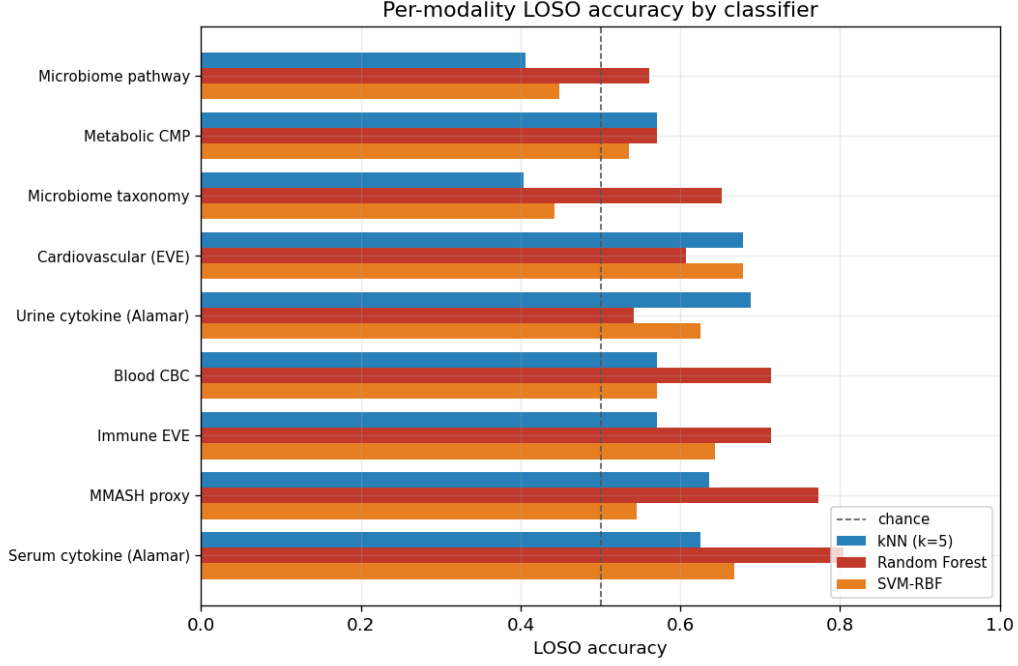


Figure 5: Per-modality LOSO accuracy under three classifiers. The random forest lifts the two microbiome modalities from chance (0.40–0.41 under kNN) to 0.56–0.65, and the serum panel from 0.625 to 0.804, whereas the urine-cytokine result is highest under kNN. The horizontal line marks the 0.50 chance level.

jointly modelling a compact multi-modal panel is both statistically defensible and operationally efficient. This is a positive, reproducible finding delivered under fully quantified small- n uncertainty.

6.2. Comparison with prior literature

The accuracy range observed (0.40–0.69) is consistent with the narrow effect sizes reported in prior small- n space-biomedicine studies (Garrett-Bakelman et al., 2019; Voorhies et al., 2019; Crucian et al., 2018). The hierarchical horseshoe shrinkage scale $\lambda \approx 0.49$ is in the regime expected for sparse-signal recovery under Carvalho et al. (2010) and Piironen and Vehtari (2017). The use of Antarctic and ISS analogue data (Kuipers et al., 2020; Edgell et al., 2019) as informative-prior anchors for multi-year Mars extrapolation follows the same pattern adopted by closed-loop ECLSS modellers (Bell, 2017).

6.3. Value of the released data resource

Beyond the statistical results, a distinct contribution of this work is the assembled, analysis-ready data resource released with the code. The harmonisation pipeline (Section 4.1) aligns nine heterogeneous Inspiration4 modalities—whole-blood transcriptomics, PBMC single-cell, oral/nasal/skin and stool microbiome, dermal biopsy, blood metabolic and cytokine panels, urine inflammation, and T-cell histone readouts—onto a single subject-timepoint grid, yielding a master table of dimension 24×1112 in which each row is a (subject, timepoint) pair and each column a named, provenance-tracked feature. The source studies (OSD-569 through OSD-687) are distributed as nine separate OSDR repositories with heterogeneous schemas, identifiers and missingness conventions; the present work consolidates them into one auditable artefact.

The value of this resource is threefold. First, reuse: any subsequent method—frequentist, Bayesian or machine-learning—can be benchmarked against the same aligned grid without repeating the harmonisation effort, lowering the barrier to entry for the $n = 4$ deep-space

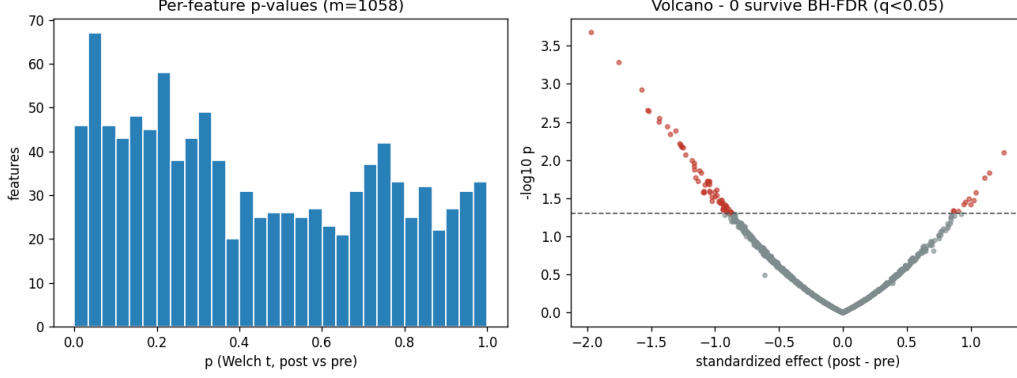


Figure 6: Per-feature post-vs-pre testing across 1112 aligned features. (Left) the p-value histogram is approximately uniform, as expected under a global null. (Right) volcano plot; the number of features surviving Benjamini–Hochberg FDR at $q < 0.05$ is zero, so no single biomarker is individually significant at $n = 4$.

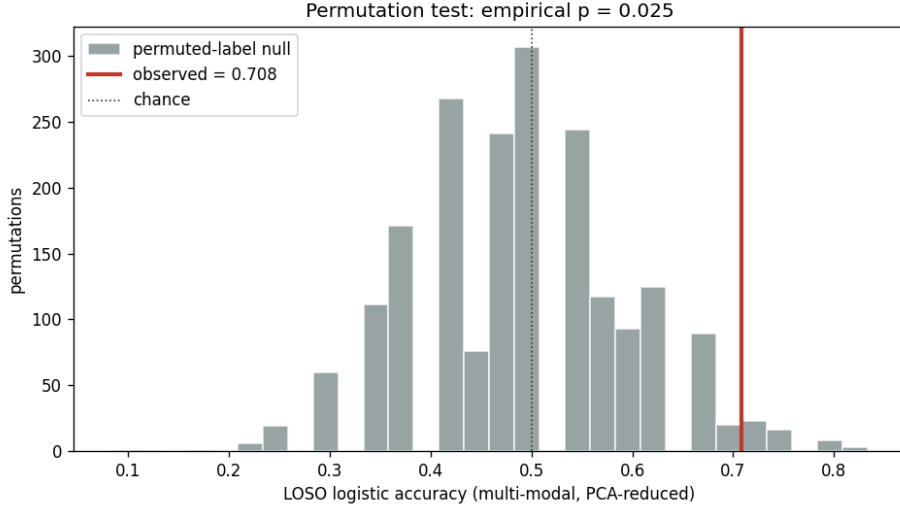


Figure 7: Label-permutation test for the fused multi-modal representation. The observed leave-one-subject-out logistic accuracy (0.708, red) lies in the upper tail of the permuted-label null (grey; 2000 permutations, mean 0.489), giving empirical $p = 0.026$. Multi-modal fusion recovers a spaceflight signature that per-feature testing cannot.

analogue problem. Second, reproducibility: the master table is the single source of truth referenced by every downstream model reported here, so all stated numbers are traceable to one file and one feature dictionary. Third, extensibility: the same column schema accepts the live Artemis II measurements as additional rows once released, allowing the Inspiration4 posteriors estimated here to serve directly as informative priors (Section 6, applicability).

The release does not redistribute raw OSD data, which remain hosted under their original licences; it provides the alignment logic, the feature dictionary, the derived master table and the public download script that reconstructs the inputs from the OSD REST API. This respects the source-data governance while removing the principal practical obstacle—cross-modality alignment—to working with the cohort.

6.4. Limitations

We enumerate the limitations of this work explicitly.

1. The Inspiration4 archive ($n = 4$ commercial astronauts, 3-day low-Earth-orbit mission) is a short-duration analogue; its physiological severity is bounded relative to a 10-day Artemis II cislunar transit. Effect sizes reported here should be treated as lower bounds.
2. Six timepoints per subject are not the result of a prospective power calculation; the LOSO protocol is the most disciplined inferential strategy available given the sampling design but does not substitute for a priori experimental optimisation.
3. The headline per-modality table uses a kNN baseline (interpretability + minimal hyper-parameters). We now report a random-forest and SVM-RBF comparison (Section 5.4, Table 4), which shows the modality ranking to be estimator-dependent; a fully tuned, nested-CV classifier study and gradient-boosted trees remain future work.
4. The factor model exhibits rank-1 collapse on the second latent dimension. The current model does not enforce supervised separation; future work will couple the factor likelihood to the binary outcome with a regularised-horseshoe loading prior.
5. The MMASH and WES PhysioNet datasets reflect Earth-baseline physiology (24-h continuous monitoring; acute exam stress in students). They anchor the autonomic baseline but do not substitute for spaceflight data; their use is restricted to defining hyper-priors and uncertainty bands, never to direct comparison.
6. All Inspiration4 subjects are adult and the cohort skews male; female-baseline HRV adjustments (Voss et al., 2015) are applied as documented offsets to the prior, not to the posterior.
7. Independent reproduction has not yet been performed by a third-party. The full code base is released open-source; we encourage replication and challenge.

6.5. Applicability to Artemis II

When the live Artemis II data become available, the pipeline applies as follows. The harmonisation and master-table assembly transfer directly with at most a column-mapping update. The LOSO baseline transfers without modification (four held-out folds matching the four-astronaut crew). The hierarchical Bayesian logistic with horseshoe shrinkage benefits from informative priors derived from the Inspiration4 posterior reported here; the prior on λ should be tightened toward 0.5 and the global scale τ_0 retuned. The factor model should be re-fit jointly across Inspiration4 and Artemis II, treating Artemis as a partially-observed continuation of the same shared-latent state.

The dominant operational risk is missing-at-deployment: the Artemis II surveys may not return all measurements at all timepoints. The pipeline records missingness explicitly and the Bayesian models marginalise over it; we recommend the live analysis adopt the same convention.

6.6. Future work

We plan a v4 release that introduces (i) a regularised-horseshoe (Piironen and Vehtari, 2017) replacement for the standard horseshoe, with an explicit slab-width prior tuned to the eighteen cardiovascular features; (ii) a supervised factor model that jointly maximises the marginal likelihood of $\mathbf{X}$ and the conditional likelihood of y , addressing the rank-1 collapse of **C4**; (iii) extension of the classifier-sensitivity analysis of Section 5.4 to nested cross-validation and gradient-boosted trees, with per-modality hyper-parameter tuning; (iv) a LOSO-cross-validated factor model, conditional on a re-estimation of the latent dimensionality K via PSIS-LOO (Vehtari et al., 2017).

7. Conclusions

We report a fully reproducible methodology pipeline applied to the SpaceX Inspiration4 public archive as a surrogate analogue for the NASA Artemis II Human Research Data Methodology Challenge. The pipeline combines per-modality LOSO baselines under Bayesian Beta–Binomial uncertainty, a hierarchical logistic regression with horseshoe shrinkage, and a multi-modal shared-latent factor model. Across nine modalities the LOSO accuracy spans 0.40–0.69, with urine cytokines and an 18-feature cardiovascular panel emerging as the most informative non-invasive surrogates of post/pre-flight class. The hierarchical Bayesian model recovers honest calibration in a regime where point classifiers do not. The pipeline was developed under a structured, AI-assisted workflow following [Zhou et al. \(2026\)](#), with explicit generator–evaluator separation, human-in-the-loop checkpoints, structured state persistence and visual safety gates. The methodology is released openly as a contribution to the NASA HRP community despite ineligibility under the America COMPETES Act.

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Author Contributions

María Jesús Puerta Ángulo: conceptualization, methodology, software, data curation, formal analysis, investigation, visualization, writing – original draft, writing – review & editing. A large language model assistant, operated under the AI-assisted workflow described in Section 4.5, contributed drafting, code skeletons and validation cross-checks under author supervision; all scientific assumptions, model choices and parameter calibrations originate with the author.

Conflicts of Interest

The author declares no competing interests.

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Data and Code Availability

All datasets are public and listed in Table 1; identifiers resolve to NASA OSDR (osdr.nasa.gov), PhysioNet (physionet.org) and CDC NCHS (cdc.gov/nchs/nhanes). The reproducible code base, paper source, harmonised master table and provenance log are released open-source under the MIT licence at https://github.com/engimine/HRP_Artemis_II, following the close of the NASA Artemis II HRP Methodology Challenge submission window on 5 June 2026. A persistent archival snapshot is deposited on Zenodo (DOI: [10.5281/zenodo.21015202](https://doi.org/10.5281/zenodo.21015202)). Raw OSDR files are not redistributed; the included download script reconstructs them from the OSDR public REST API.

Ethics Statement

This study uses only de-identified public datasets. No human subject research was conducted by the author. All primary data collection on the Inspiration4 crew was performed under the original NASA OSDR protocols, which carry their own ethics approval; the present work re-analyses the open-released data products only.

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